

GWT Cheat Sheet

GWT 2.13.1 · gwt-maven-plugin 1.3.0 · JDK 21, -sourceLevel 17 · Jakarta Servlet 6 · verified 2026-10-08

Setup and dev loop

BOM `org.gwtproject.gwt` (pom, import): `gwt-user`, `gwt-dev`, `gwt-codeserver`, `gwt-servlet-jakarta`, `jsinterop-annotations`. `RequestFactory` is `org.gwtproject.web.binder.*`.

Plugin `net.ltgt.gwt.maven:gwt-maven-plugin`; `packagings gwt-app`, `gwt-lib`; `goals compile, codeserver, devmode, test, generate-module, package-app/lib, add-super-sources`.

Archetype `mvn archetype:generate -DarchetypeGroupId=net.ltgt.gwt.archetypes -DarchetypeArtifactId=modular-webapp -DarchetypeVersion=2026.6.22 -Dmodule-short-name=bookshelf -> client / shared / server`.
`mvn gwt:codeserver -pl bookshelf-client -am (SDM, :9876)`
`mvn jetty:run -pl bookshelf-server -am -Denv=dev (:8080)`
`mvn clean package = production compile + war. -pl *-client fails on Maven 3.9.`

Module XML (*.gwt.xml)

```
<module rename-to><inherits name> (User, Core, I18N, HTTP, JSON, XML, Logging, Resources), <entry-point class>, <source path>, <public path>, <super-source>, <stylesheet src>, <script src>, <servlet> (embedded server only), <set-property>, <set-configuration-property>, <add-linker>, <collapse-property>.
```

Properties `user.agent = gecko1_8, safari only (2.10+); locale; compiler.stackMode; gwt.logging.*; CssResource.enableGss; jre.checks.checkLevel`.
Host page: `<script src="bookshelf/bookshelf.nocache.js">`, `<meta name="gwt:property" content="locale=es">`.

Compiler and code server

`com.google.gwt.dev.Compiler: -war -deploy -extra -workDir -gen, -style OBFUSCATED|PRETTY|DETAILED, -optimize 0-9, -draftCompile, -sourceLevel 17 (CLI default 1.8!), -failOnError (=strict), -validateOnly, -localWorkers, -incremental, -setProperty k=v, -checkAssertions, -saveSource, -compileReport, -logLevel, -generateJsInteropExports (+ include/exclude regex).`

CodeServer `-port 9876 -src -launcherDir -workDir -bindAddress -[no]incremental -[no]precompile -style -setProperty -[no]failOnError -sourceLevel`. No auth, serves sources: bind to localhost. SDM default since 2.7.

JRE emulation

Supported: most of `java.lang`, `java.util` (streams, Optional), `java.util.logging`, `java.sql` `Date/Time/TimeStamp`. Missing: reflection, threads, `java.io.File`, `java.time`, `java.util.regex`, `UUID`, `java.net`, `String.format`. Java 8-17 syntax with `-sourceLevel`; records need custom RPC serializers. Long is emulated (slow), int overflow wraps, `String.split`. Java 8 rules; dates: `java.util.Date + DateFormat`. Shared code: `super-source`, `GWT.isClient()`, `GWT.isScript()`, `GWT.getModuleBaseURL()`, `GWT.log`.

JsInterop (and JSNI mapping)

`@JsType(isNative=true, namespace=JsPackage.GLOBAL, name="X");`
`@JsMethod @JsProperty @JsConstructor`
`@JsFunction @JsOverlay @JsIgnore @JsOptional` (`@JsNullable`/`@JsNonNull`/`@JsAsync`/`@JsEnum` do nothing in 2.13.1). Export to JS needs `-generateJsInteropExports`. `jsinterop-base`: `Js`, `JsPropertyMap`, `Any`. `Elemental2 1.3.2` (dom, core, promise, webstorage).

JSNI → **JsInterop**: native method + `$wnd/$doc` → native `@JsType` on `JsPackage.GLOBAL`; `JavaScriptObject` overlay → `@JsType(isNative)` + `@JsOverlay`; `Java` callback → `@JsFunction`. JSNI is ES5 only.

Deferred binding

`GWT.create(X.class)` resolved at compile time per permutation. `<replace-with class>` / `<generate-with class>` with `<when-type-is>`, `<when-type-assignable>`, `<when-property-is>`, `<all>`, `<any>`, `<none>` (= NOR). Last declared rule examined first. `<define-property>`, `<extend-property>`, `<property-provider>`, `<set-property-fallback>`. Generators: `Generator/IncrementalGenerator`, `TypeOracle`, `SourceWriter`; avoid for new code.

Widgets, panels, handlers

Widgets `Label`, `HTML`, `Button`, `PushButton`, `ToggleButton`, `CheckBox`, `RadioButton`, `TextBox`, `PasswordTextBox`, `TextArea`, `ListBox`, `SuggestBox`, `DatePicker`, `Tree`, `MenuBar`, `TabBar`, `Image`, `Anchor`, `Hyperlink`, `FileUpload`, `Hidden`. `RootPanel.get()`, `RootLayoutPanel.get()`.

Basic panels `FlowPanel`, `SimplePanel`, `HTMLPanel`, `ScrollPanel`, `FocusPanel`, `FormPanel`, `Grid`, `FlexTable`, `DisclosurePanel`, `PopupPanel`, `DialogBox`. **Layout panels** `LayoutPanel`, `DockLayoutPanel`, `SplitLayoutPanel`, `TabLayoutPanel`, `StackLayoutPanel`, `DeckLayoutPanel` (standards mode; `ProvidesResize/RequiresResize`).

Handlers `addClickHandler(e -> ...)` returns `HandlerRegistration` (`removeHandler()`); `sinkEvents + onBrowserEvent`; `Event.addNativePreviewHandler`; `SimpleEventBus`. Listeners are deprecated. No pointer/input types: `addBitlessDomHandler`. Use `setText`, not `setHTML`, for user data.

Scheduler `scheduleDeferred`, `scheduleEntry`, `scheduleFinally`, `scheduleFixedDelay`, `scheduleFixedPeriod`, `scheduleIncremental`; `Timer.schedule/scheduleRepeating/cancel`. `DeferredCommand` deprecated.

UiBinder, cell widgets, Editors

UiBinder `Foo.java + Foo.ui.xml`; interface `Binder` extends `UiBinder<Widget, Foo>`; `GWT.create`; `@UiField [(provided=true)]`, `@UiHandler("id")`, `@UiFactory`, `@UiConstructor`, `@UiChild`, `@UiTemplate`; elements `ui:style ui:with ui:image ui:data ui:import ui:msg ui:ph ui:attribute ui:safehtml ui:text; UiRenderer` (interface) for cells.

Cells `CellList` (page 25), `CellTable` (15), `DataGrid` (50), `CellTree`, `CellBrowser`; `Column`, `AbstractCell + SafeHtmlTemplates`, `FieldUpdater`; `ListDataProvider`, `AsyncDataProvider` (Range); `ListHandler/AsyncHandler` sorting; `SimplePager`; `SingleSelectionModel`, `MultiSelectionModel`, `ProvidesKey`.

Editors `SimpleBeanEditorDriver` (initialize, edit, flush, hasErrors); `Editor<T>`, `IsEditor`, `LeafValueEditor`, `ValueAwareEditor`, `CompositeEditor`, `HasEditorErrors`; `@Path`, `@Ignore`; `ListEditor`, `ValueBoxEditorDecorator`.

ClientBundle, CSS and GSS

interface `Res` extends `ClientBundle`, `GWT.create`, `@Source("x.png")`. Types: `ImageResource` (`@ImageOptions`), `DataResource` (`@MimeType`), `TextResource`, `ExternalTextResource` (needs `eval/CSP unsafe-eval`), `CssResource`, `GwtCreateResource`. Inlining on (data URLs, ~48 KiB). `ensureInjected()`. `ImageBundle` is deprecated.

GSS `CssResource.enableGss=true`, `conversionMode` (off default, strict while migrating), `gssDefaultInUiBinder` (all default off); directives `@def` `if/else` `eval` `external` `sprite` `url`; no `var()` nesting. Themes `Clean`, `Standard`, `Chrome`, `Dark`. `@Shared` does not cross bundles.

Internationalization

Constants, `ConstantsWithLookup`, `Messages` (`@DefaultMessage` `@Key` `@Description` `@PluralCount` `@AlternateMessage` `@Select` `@Offset`); `Dictionary`; `LocaleInfo`; `NumberFormat`, `DateFormat`. Properties `Msgs_es`. properties next to the interface; `<extend-property name="locale" values="es"/>`; select with `?locale=es`. A single apostrophe in `Messages` is a compile error: write `' '`. `i18nCreator` deprecated.

History, Activities and Places

`History.newItem(token)`, `replaceItem` (2.7+), `addValueChangeHandler` then `History.fireCurrentHistoryState()`; no history `iframe` in 2.13. **MVP**: `Place`, `PlaceTokenizer`, `@Prefix`, `PlaceHistoryMapper` + `@WithTokenizers`, `PlaceController`. `goTo`, `PlaceHistoryHandler`. `register/handleCurrentHistory`, `AbstractActivity.start(panel, eventBus)`, `mayStop`, `onStop`, `ActivityMapper`, `ActivityManager`. `setDisplay`. DI: `Dagger 2.60.1` works, `GIN` is dead on JDK 21. One `GWTTestCase` module per test class.

Client-server communication

GWT-RPC interface `BookService` extends `RemoteService + @RemoteServiceRelativePath("books") + BookServiceAsync` with `AsyncCallback<T>`; server `RemoteServiceServlet` (`com.google.gwt.user.server.rpc.jakarta` for Jakarta). Types: `IsSerializable/Serializable`, default constructor, custom field serializer. Policy files `*.gwt.rpc`. Exceptions: `StatusCodeException`, `InvocationException`, `IncompatibleRemoteServiceException`, `SerializationException`. Java 17+: `--add-opens java.base/java.lang=ALL-UNNAMED`. Protocol v7 is eval'd.

RequestBuilder `new RequestBuilder(GET, url).sendRequest(null, new RequestCallback(){...})`; status 0 arrives in `onResponseReceived`; `setHeader` drops default Content-Type. **JSON**: `JsonUtils.safeEval` (= `JSON.parse`), `JSONParser.parseStrict`, `JsonRequestBuilder` (avoid), `XMLParser`. **REST**: `Elemental2` fetch, `domino-rest`, `RestyGWT`, `AutoREST`, `gwt-jackson`.

RequestFactory `RequestFactory`, `RequestContext`, `EntityProxy`, `ValueProxy`, `@ProxyFor`, `@Service`, `Receiver` (`onSuccess`, `onConstraintViolation`), `requestFactory-apt` required; `AutoBeans`: `AutoBeanFactory`, `AutoBeanCodex`.

Code splitting, linkers, caching

`GWT.runAsync(new RunAsyncCallback(){onSuccess, onFailure})`; initial / leftovers / exclusive fragments; `compiler.splitpoint.initial.sequence`; `XfragmentCount`; `async` provider pattern. `XfragmentMerge` is unknown in 2.13.1. **Linkers** `xsiframe` (default), `direct_install`, `sso` (fails with split points); `std/xs` deprecated. **Caching** `*.nocache.js`: no-cache; `*.cache.js`, `deferredjs/`: long-lived immutable; keep `.gwt.rpc` policy files deployed. Report: `-compileReport` → Story of Your Compile.

Testing

`GWTTestCase` (`getModuleName()`, `delayTestFinish(ms)`, `finishTest()`), `GWTTestSuite`; `mvn gwt:test` with `-Dgwt.args="-runStyle HtmlUnit:FF"`; run styles `HtmlUnit` (default), `Manual`, `ExternalBrowser`, `devMode`; `@DoNotRunWith(Platform.HtmlUnitLayout)`. Presenters: plain `JUnit` with mocked views (`GwtMockito` archived). `JaCoCo` sees no `GWTTestCase` coverage. **E2E**: `Playwright` for Java.

Security strip

XSS `SafeHtml`, `SafeHtmlBuilder`, `SafeHtmlUtils.htmlEscape`, `SafeHtmlTemplates`, `SafeUri`, `SafeStyles`; never `setHTML` with user data; sanitize with `OWASP Java HTML Sanitizer`. **XSRF** `XsrfProtectedService`, `RpcToken`, `XsrfTokenService` / `XsrfTokenServiceServlet`, `@XsrfProtect`; Spring: `X-XSRF-TOKEN` header. **Never expose the code server** (no auth, serves sources). **CSP** `script-src 'self'` needs a non-inline linker output; `nonce + 'strict-dynamic'` works; `RPC v7 eval` needs `unsafe-eval` or `-Dgwt.rpc.version=8`. Authorise server-side. CVE-2013-4204, 2012-5920, 2012-4563 fixed by 2.5.1; recompile.

Changed since older tutorials

listeners → handlers (1.6/2.0) · `ImageBundle` → `ClientBundle` (2.0) · hosted mode → `Super Dev Mode` (2.5, default 2.7) · `com.google.gwt` → `org.gwtproject` (2.10+) · IE removed, `user.agent` `gecko1_8/safari` (2.10) · `javax` → `jakarta` servlets (`gwt-servlet-jakarta`) · `JSNI` → `JsInterop` (2.8) · `Elemental` → `Elemental2` · `org.codehaus.mojo` plugin → `net.ltgt` · `CSS` → `GSS` · `webAppCreator` deprecated · `Dev Mode` server serves only static files (2.13) · tools need Java 11+ (2.12).